

Efficacy and Safety of Low-Dose Iodine Plaque Brachytherapy for Juxtapapillary Choroidal Melanoma



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- **PURPOSE:** To evaluate low- vs high-dose plaque brachytherapy for juxtapapillary choroidal melanoma.
- **DESIGN:** Retrospective interventional case series.
- **METHODS:** SETTING: Single institution. STUDY POPULATION: Forty-seven patients with juxtapapillary choroidal melanoma. INTERVENTION: Iodine-125 plaque brachytherapy. Eyes were divided into apex low-dose (LD) and high-dose (HD) groups ($\leq$ or $>$ median apex dose 84.35 Gy). Main outcome measures were time to distant failure, local failure, death, enucleation, radiation retinopathy, optic neuropathy, and best-corrected visual acuity (BCVA).
- **RESULTS:** Freedom from distant failure rates were 96% and 95% in apex LD and HD groups at 5 years and 77% and 95% at 10 years, respectively ($P = .84$). Freedom from local failure rates were 90% in the apex LD group vs 89% in the HD group at 5 and 10 years ($P = .96$). Apex LD and HD groups did not differ for time to death or enucleation. Five- and 10-year freedom from radiation retinopathy and optic neuropathy rates were higher in the apex LD than HD group. Loss of ≥ 3 BCVA lines, final BCVA 20/40 or better, and final BCVA 20/200 or worse were more favorable in the 5 mm LD compared to HD group. Visual acuity outcomes did not differ between apex LD and HD groups.
- **CONCLUSIONS:** Low-dose iodine-125 plaque brachytherapy (67.5–81 Gy at tumor apex) provides safe and effective tumor control for juxtapapillary choroidal melanoma and may be associated with reduced radiation

toxicity. Larger trials are needed to determine the optimal therapeutic dose for juxtapapillary choroidal melanoma. (Am J Ophthalmol 2018;186:32–40. © 2017 Elsevier Inc. All rights reserved.)

BRACHYTHERAPY FOR JUXTAPAPILLARY CHOROIDAL malignant melanoma (CMM) poses a significant treatment challenge owing to the proximity to the optic nerve and macula, as well as its posterior location, which is difficult to access surgically. Because these unique characteristics may impede optimal brachytherapy results, juxtapapillary tumors were excluded from the Collaborative Ocular Melanoma Study (COMS).¹ Historically, enucleation was often favored for juxtapapillary melanoma; however, brachytherapy can achieve equivalent survival rates with the benefit of eye preservation.² Specific brachytherapy plaque designs for tumors with optic nerve proximity have been designed and tested, including notched^{3,4} and slotted plaques.⁵

The 2014 American Brachytherapy Society – Ophthalmic Oncology Task Force guidelines indicate that brachytherapy for juxtapapillary CMM is “controversial,”⁶ owing to increased risk for radiation toxicity and sub-optimal local control demonstrated in historical reports.^{4,7} Notably, plaque brachytherapy for juxtapapillary CMM carries a particularly high risk for vision-related morbidity, including radiation retinopathy, maculopathy, and optic neuropathy.^{4,5,8,9}

Preferred radiation strategies for the treatment of juxtapapillary CMM have not been defined, but large published series have used radiation doses between 70 and 110 Gy prescribed to the tumor apex.^{3–5,8–10} More recent analyses of plaque brachytherapy have reported local control rates $>90\%$ for juxtapapillary CMM,^{5,10} in contrast to historical series with local control rates closer to 50%.^{4,7} It is not uncommon for juxtapapillary tumors to be treated with COMS-like treatment parameters, including prescribing 85 Gy to tumor apex or a minimum 5 mm height. At our institution, a range of prescription doses, including doses below 85 Gy to the tumor apex, have been used over the past 2 decades.

Two studies of plaque brachytherapy for CMM have demonstrated decreased radiation complications and

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